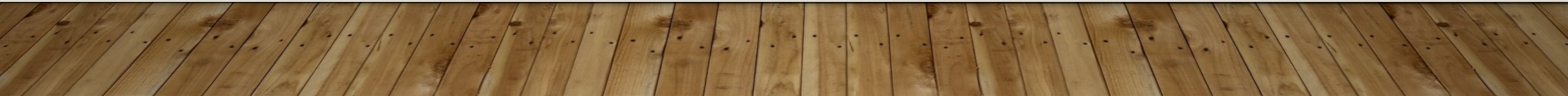


BIO262- PHYSIOLOGY

MODULE 7

EYES

EARS



VISIONS



Armenia

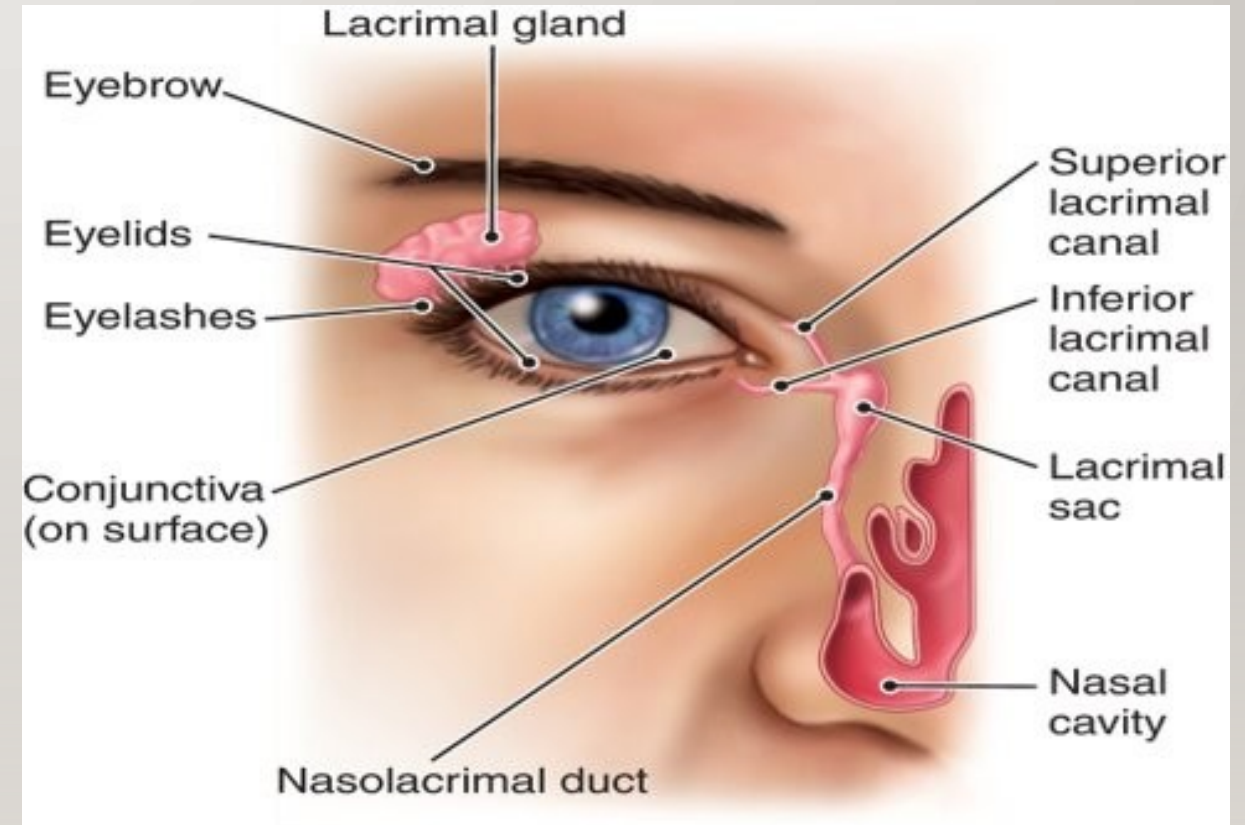
Beauty is in the beheld eye of a 16-year-old boy in Yerevan. In this highly magnified view of his iris's surface architecture, the central black pool is the pupil, and his lashes are reflected by the cornea. His eyelids appear as pink rims at top and bottom.

PHOTO: SUREN MANUELYAN

TEARS

Why do we have tears?

- ✓ Helps wash away debris
- ✓ Lubricates the cornea
- ✓ Supplies O₂/nutrients to cornea



CHAMBERS OF THE EYE

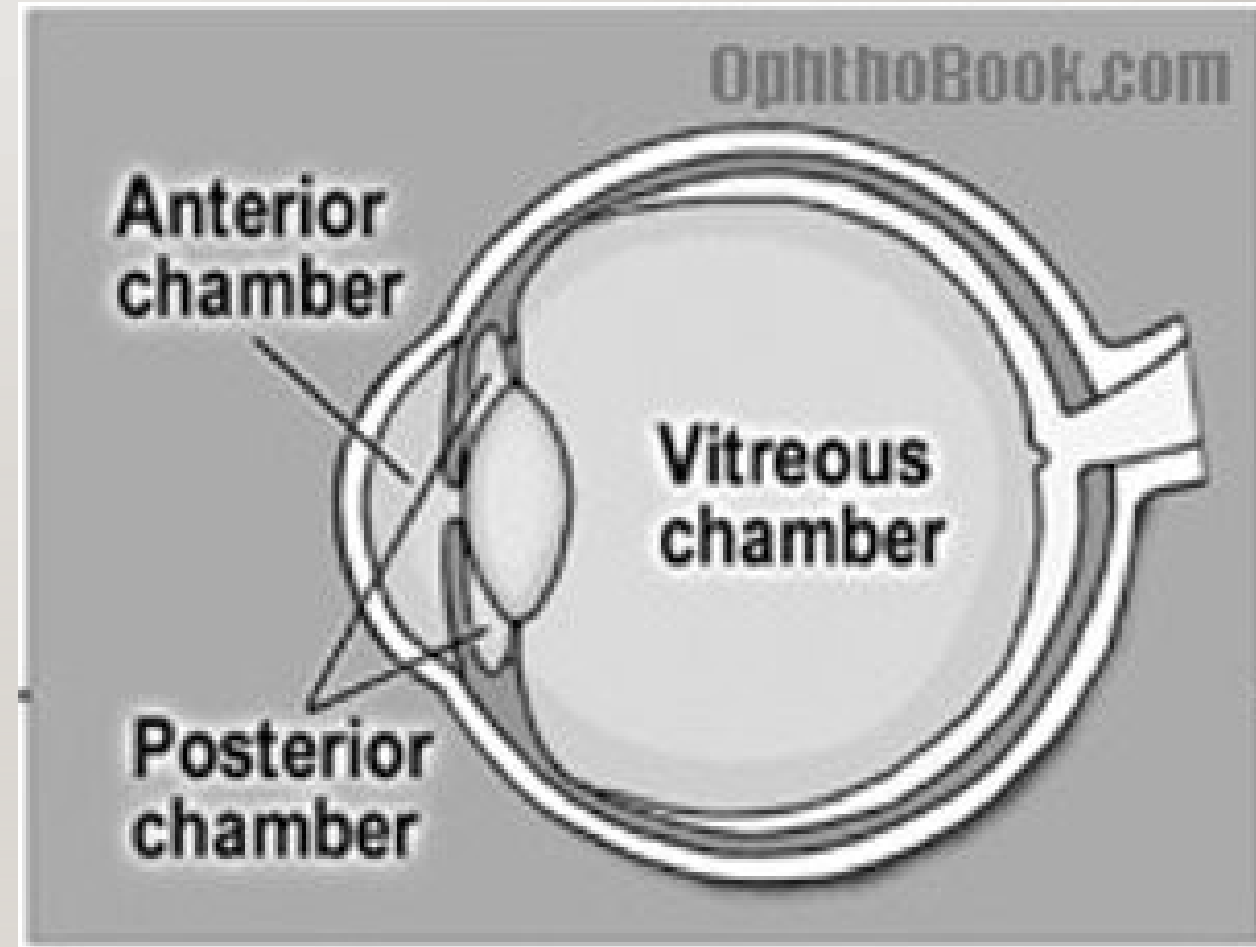
Vitreous humor:

Very gel-like and never replaced
- Keeps the retina in place and helps maintain the shape of the eye.

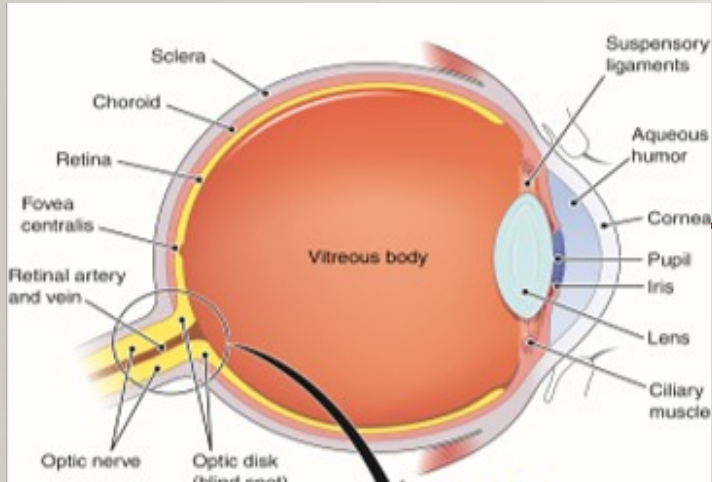
Aqueous humor:

More watery - and continuously replaced.

- **Maintains intraocular pressure.**
(measured to monitor for glaucoma)

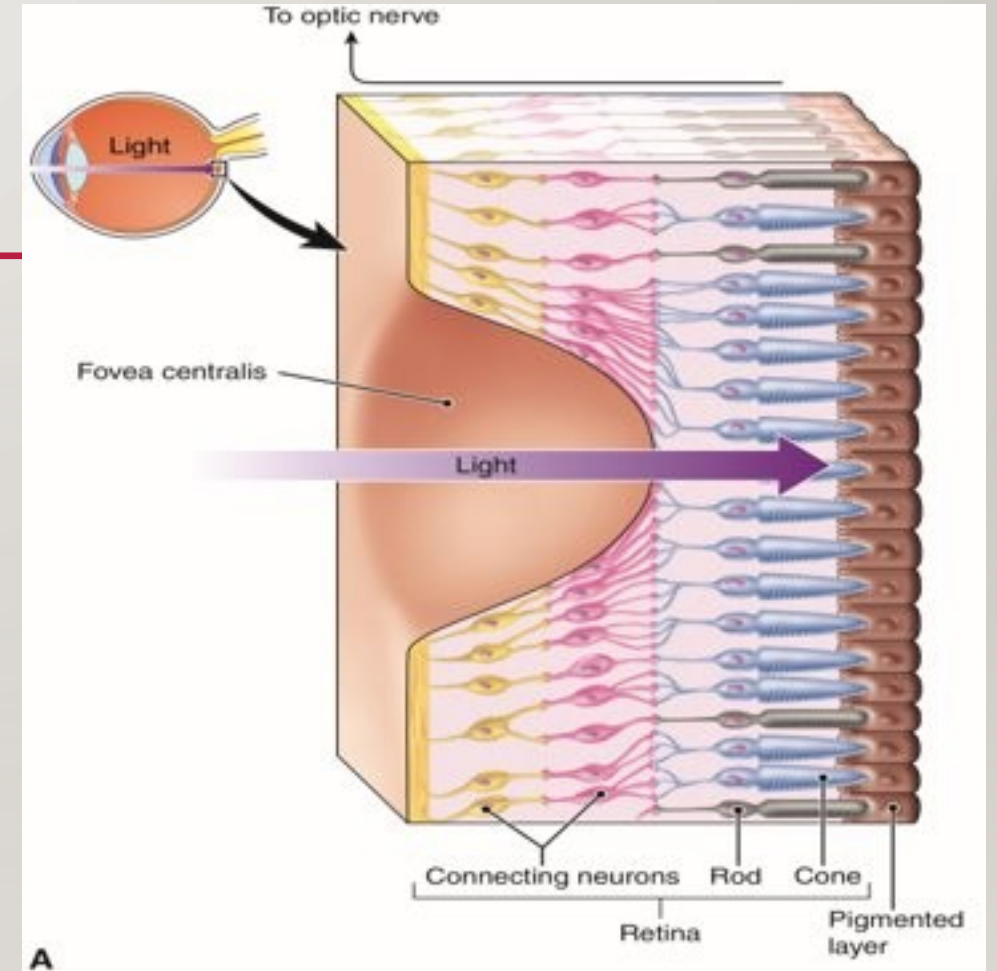


LAYERS OF THE EYE



Retina – changes light to electrical signals
Light hits rods/cones these are connected to bipolar cells that are connected to ganglion cells which transmit visual information to the optic nerve.

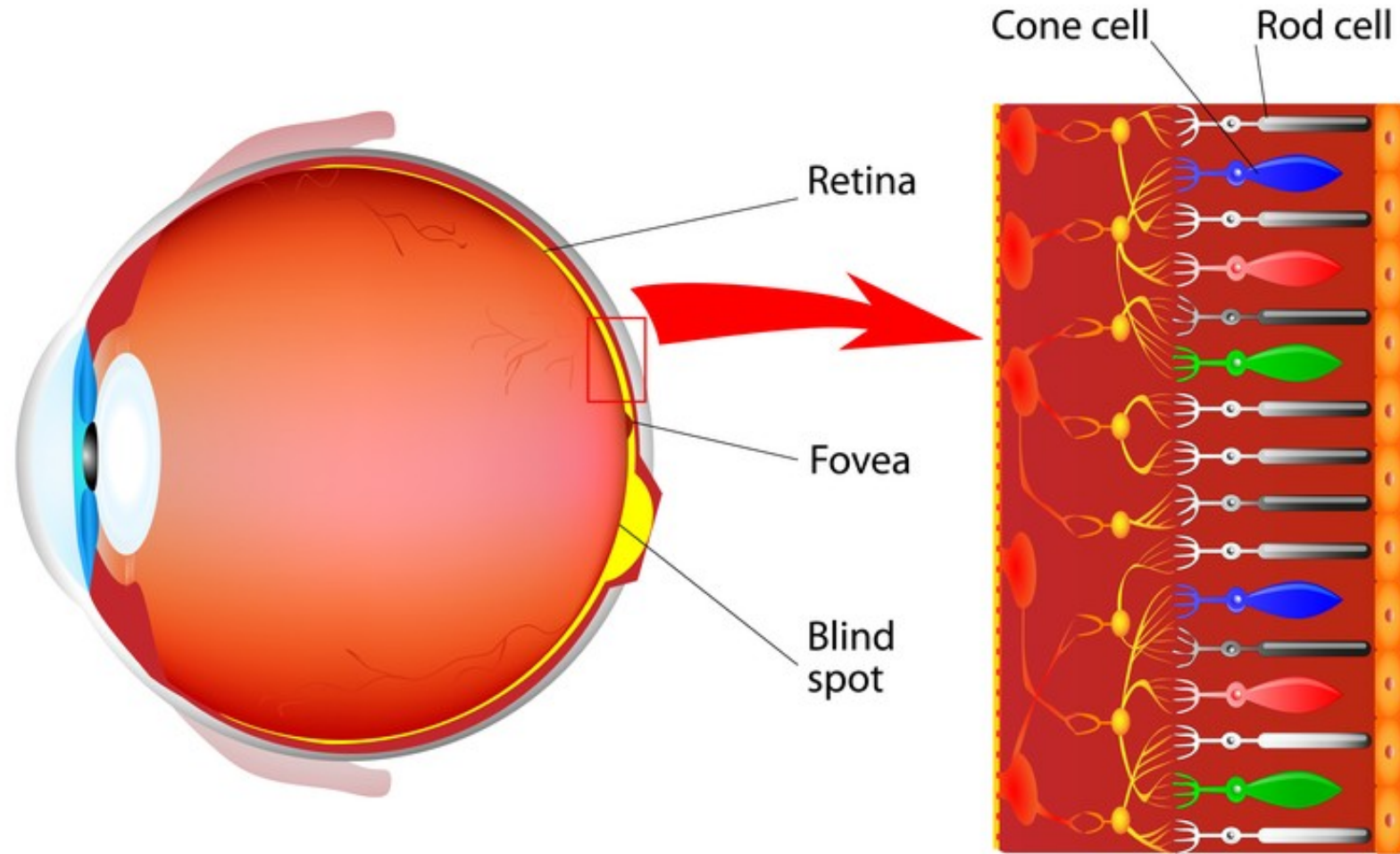
Optic nerve sends information to the visual cortex in the occipital lobe.



Rods/cones

What
triggers
depolarization
of these
neurons?

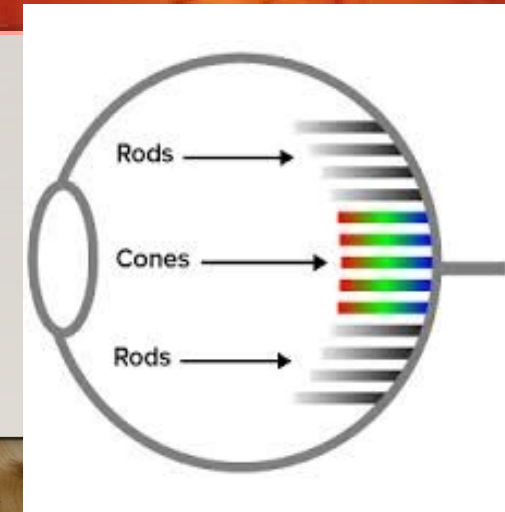
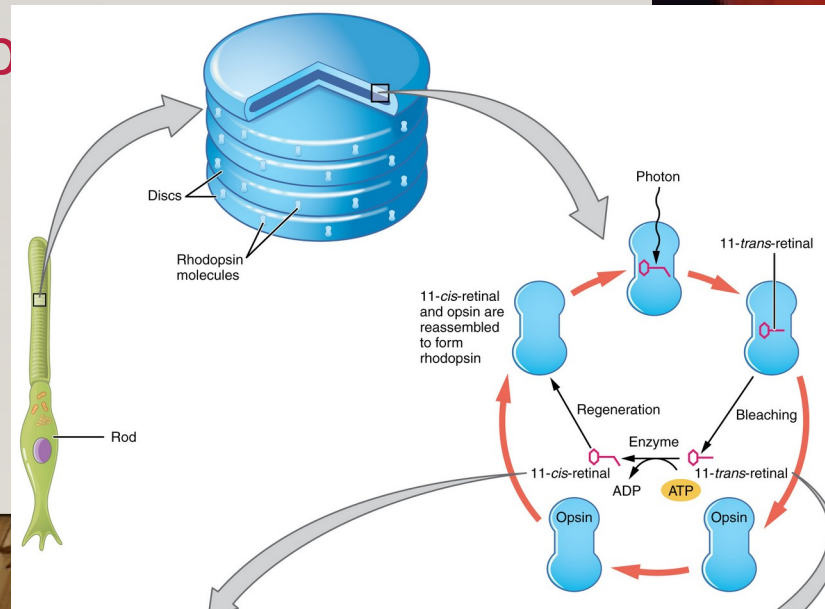
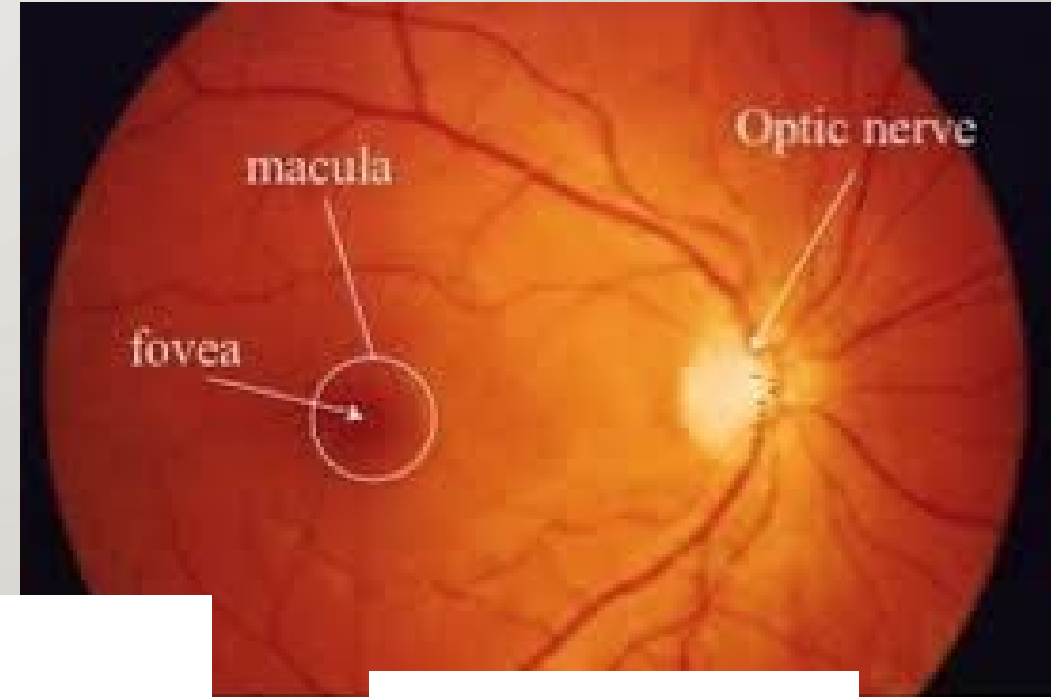
Photoreceptor cell



VISUAL PIGMENTS & PHOTORECEPTION

Rhodopsin—visual pigment in the photoreceptors comprised of **opsin** **bound to retinal** (comes apart when light hits it)

- Retinal— same in rods/cones
- Opsin- cones have three different types - Triggers a resp



REFRACTION

▶ Bending of light so it hits the retina.

▶ Two main structures:

▶ Cornea

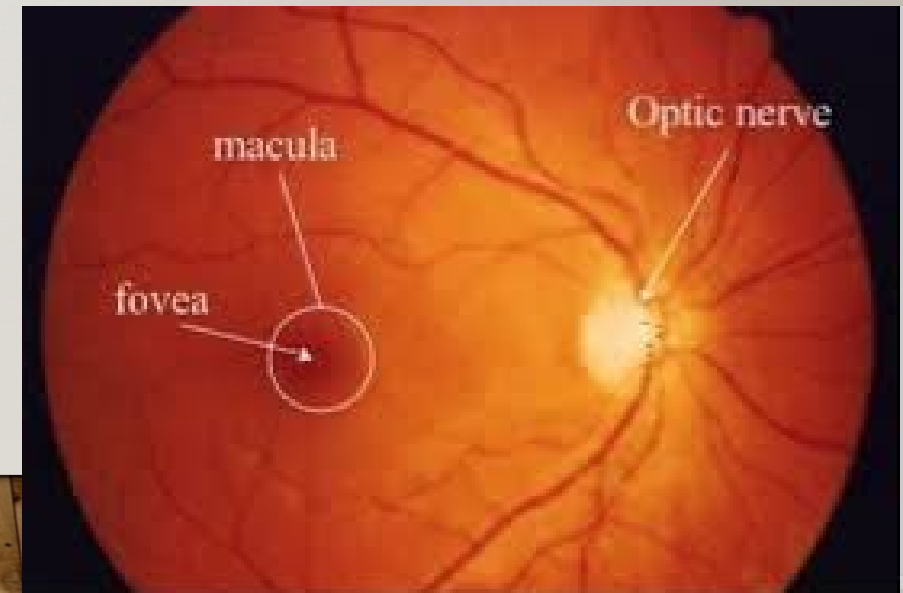
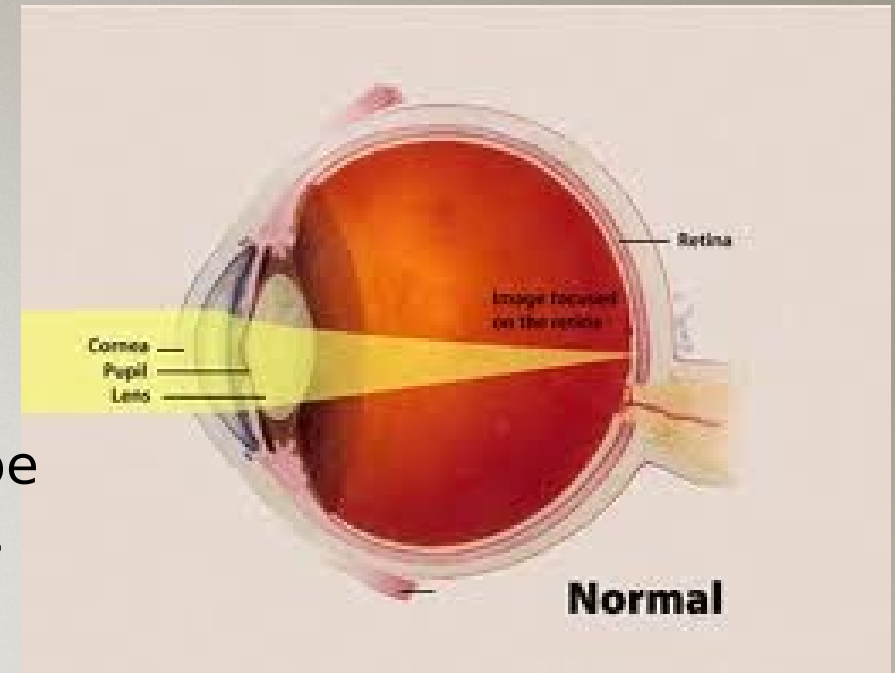
▶ Lens

- **Accommodation** – The lens changes shape based on whether you are looking far away/distance, or near.

- FLATTENS when you look in the distance (flat like a roadway)

- ROUNDS when you look near (round like the ring on your hand)

- https://www.youtube.com/watch?v=fXd3L4izfQI&feature=emb_title



PRACTICE QUESTION

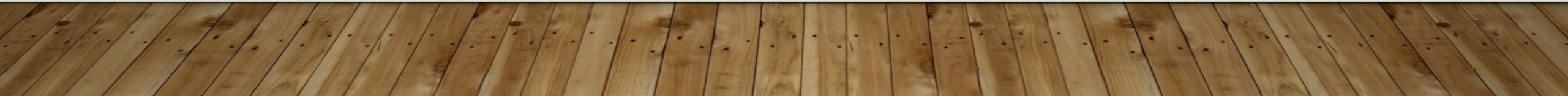
- Joseph is on a walk and is looking in the distance at the mountains. He suddenly gets a text message and looks down at his phone to read it. Which would be most likely to happen when he looks down at his phone?

A – his cornea will retract

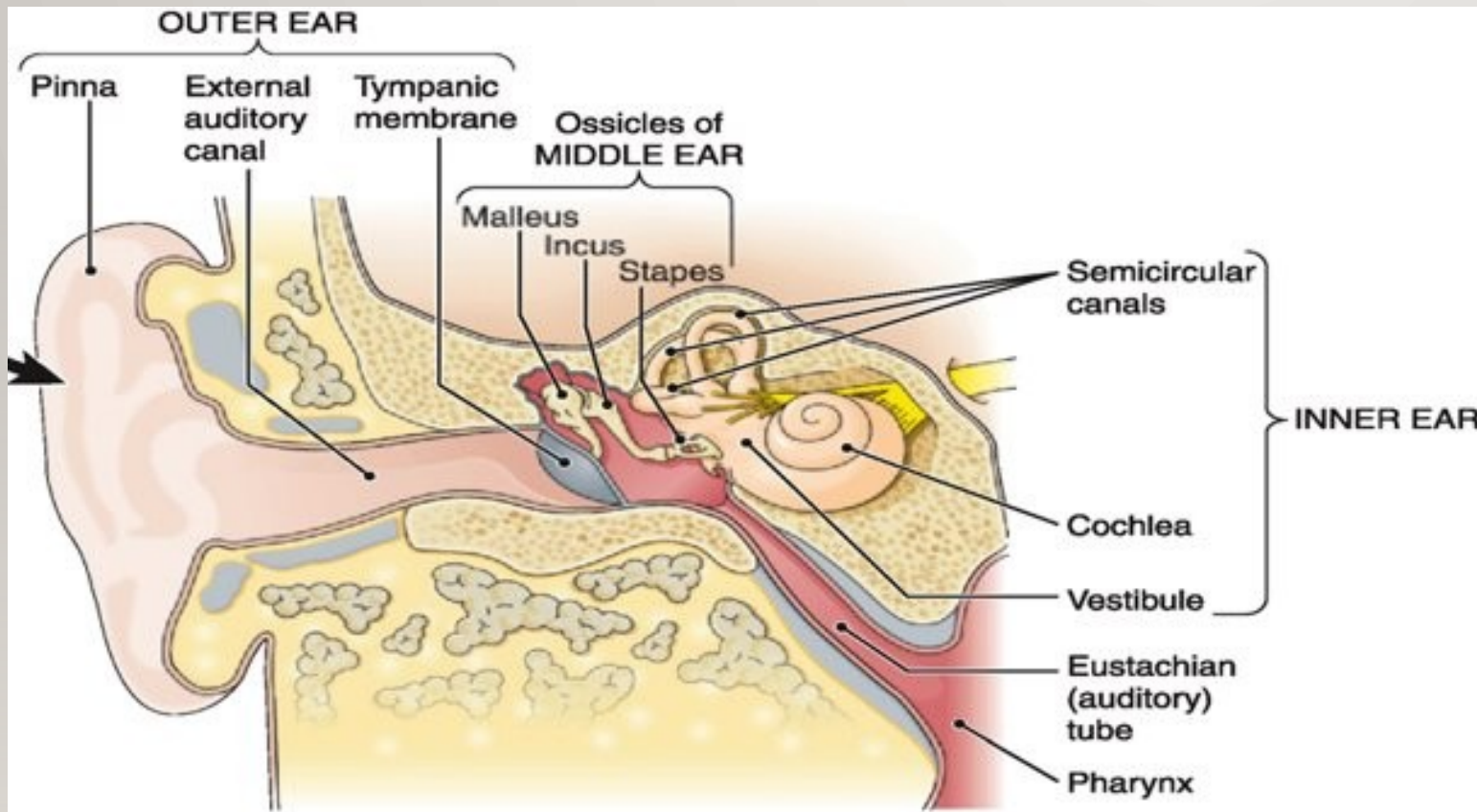
B – the ciliary bodies will change the shape of the lens so that it is more rounded

C – the pupil will dilate via relaxation of the smooth muscle of the iris

D – the ciliary bodies will change the shape of the lens so that it is more flat



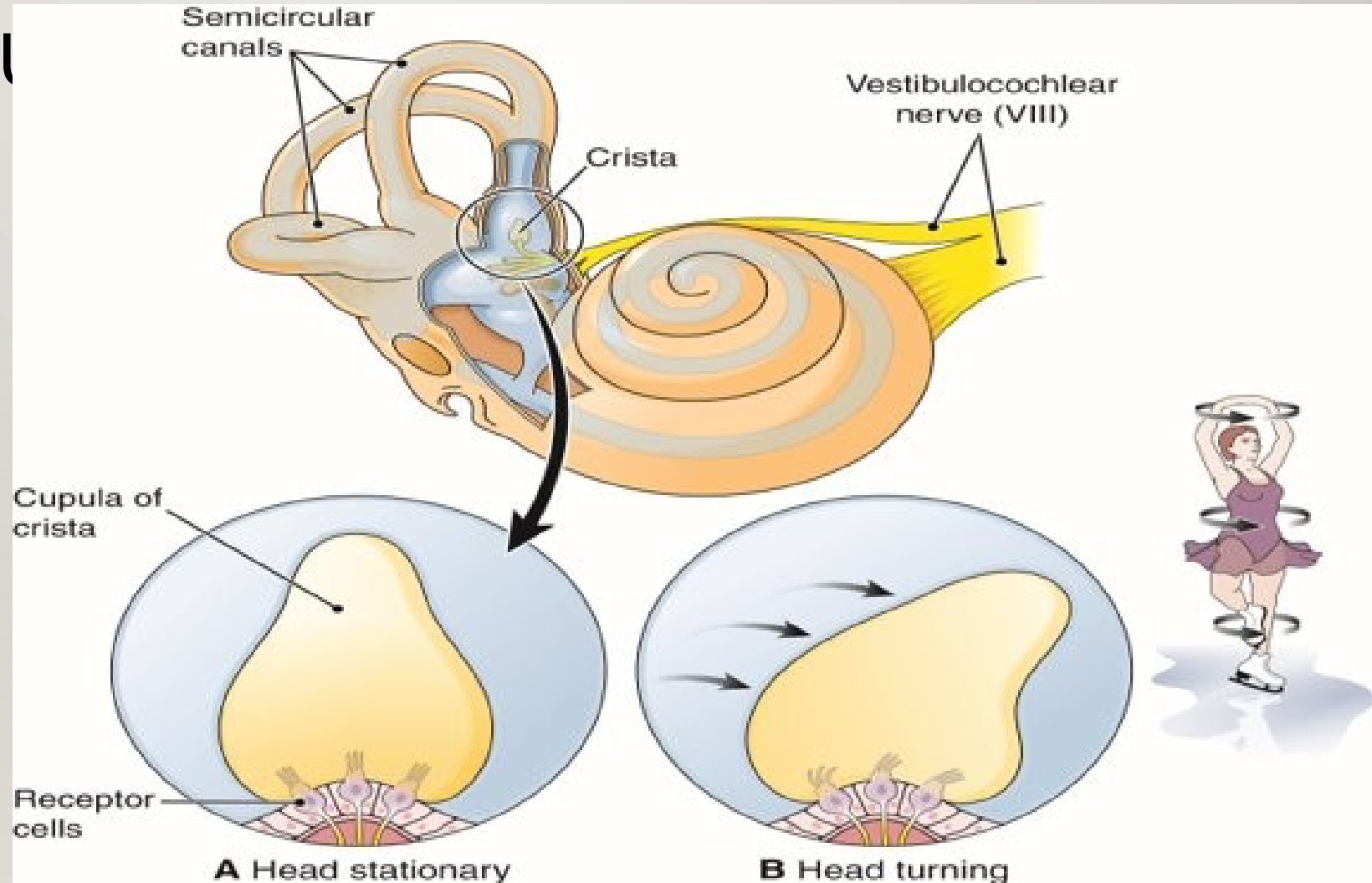
STRUCTURE OF THE EAR



- External Ear:
 - Pinna captures sound waves
 - Tympanic membrane vibrates ossicles
- Middle Ear:
 - Ossicles amplify sound and vibrate oval window.
 - Eustachian tube equalizes pressure in ear.
- Internal Ear:

SEMICIRCULAR CANALS OF THE VESTIBUL

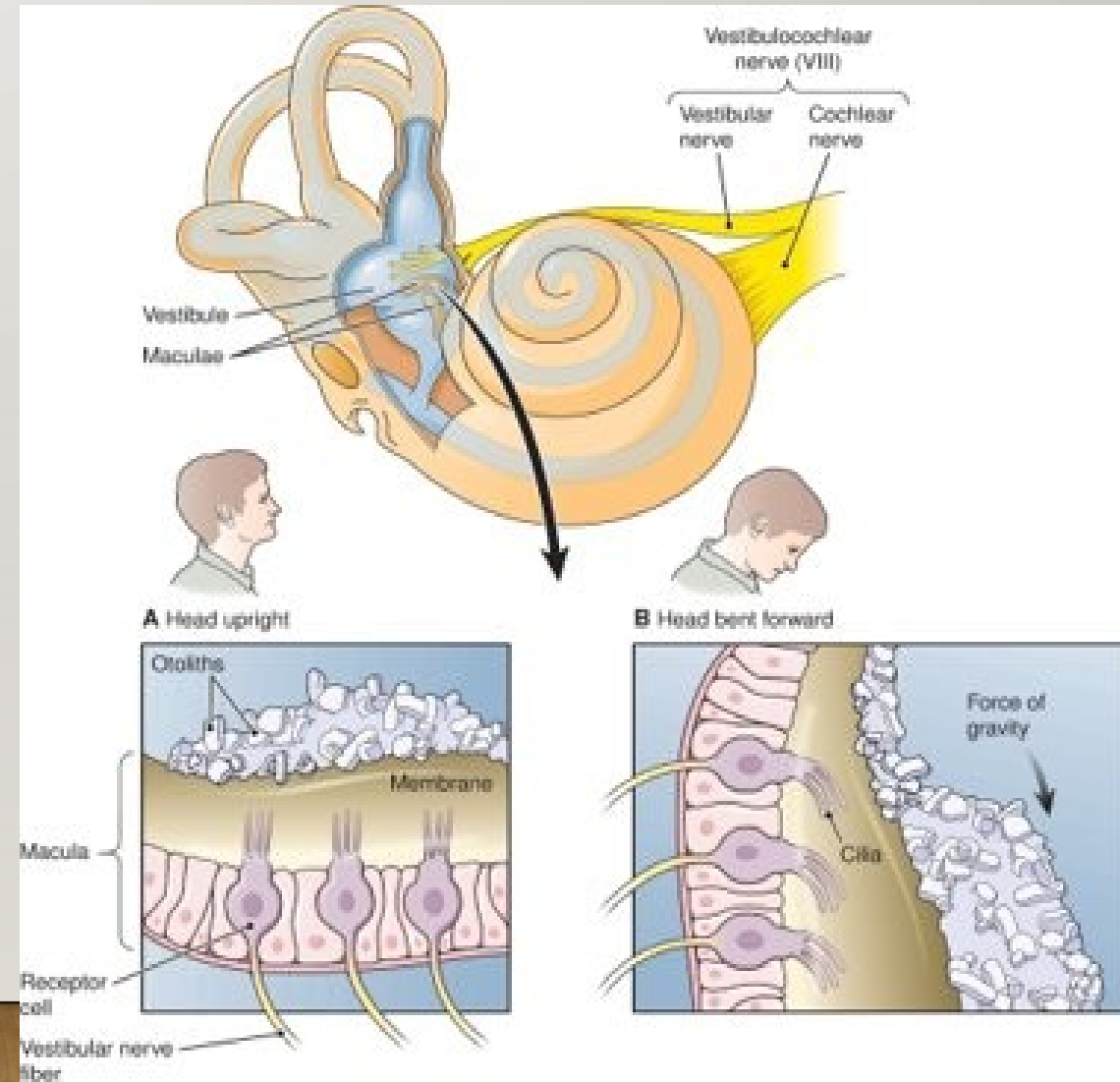
What do these
do?
Which type of
movement do
they detect?
How?



VESTIBULE - UTRICLE/SACCCULE

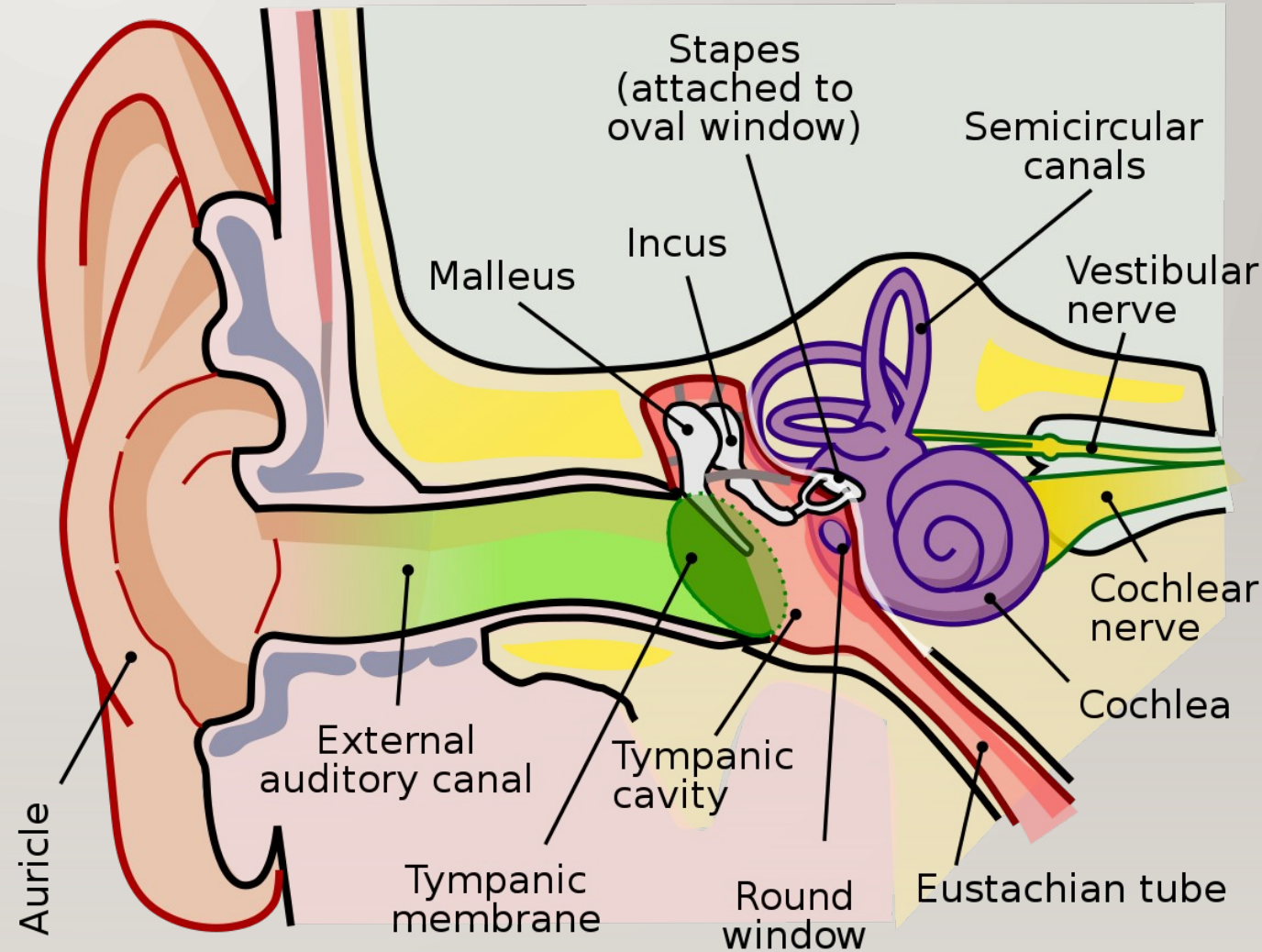
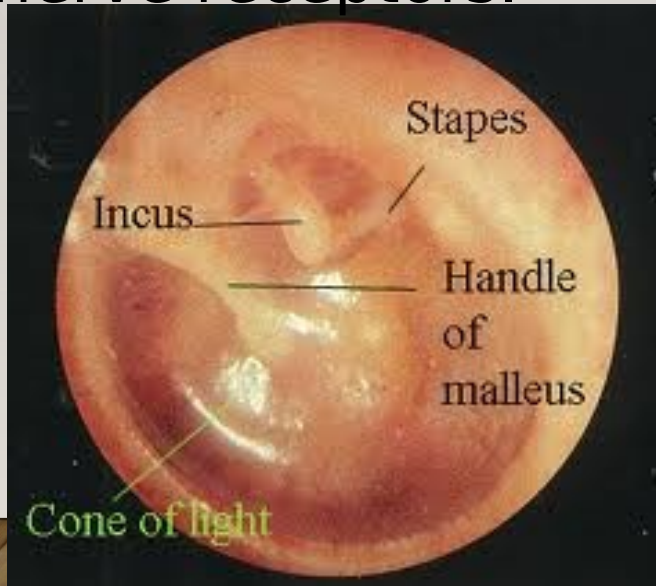
What do these do?
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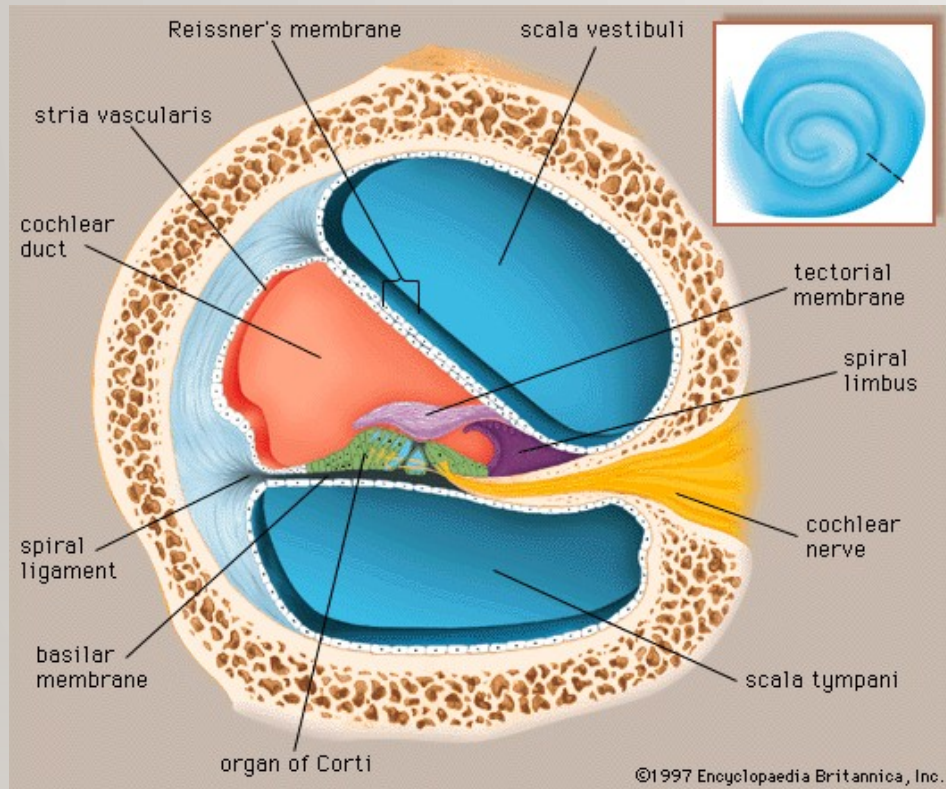
Vestibular structures
communicate with the
Cerebellum which coordinate
messages from joints, muscles,
eyes, ears to coordinate and
maintain balance.



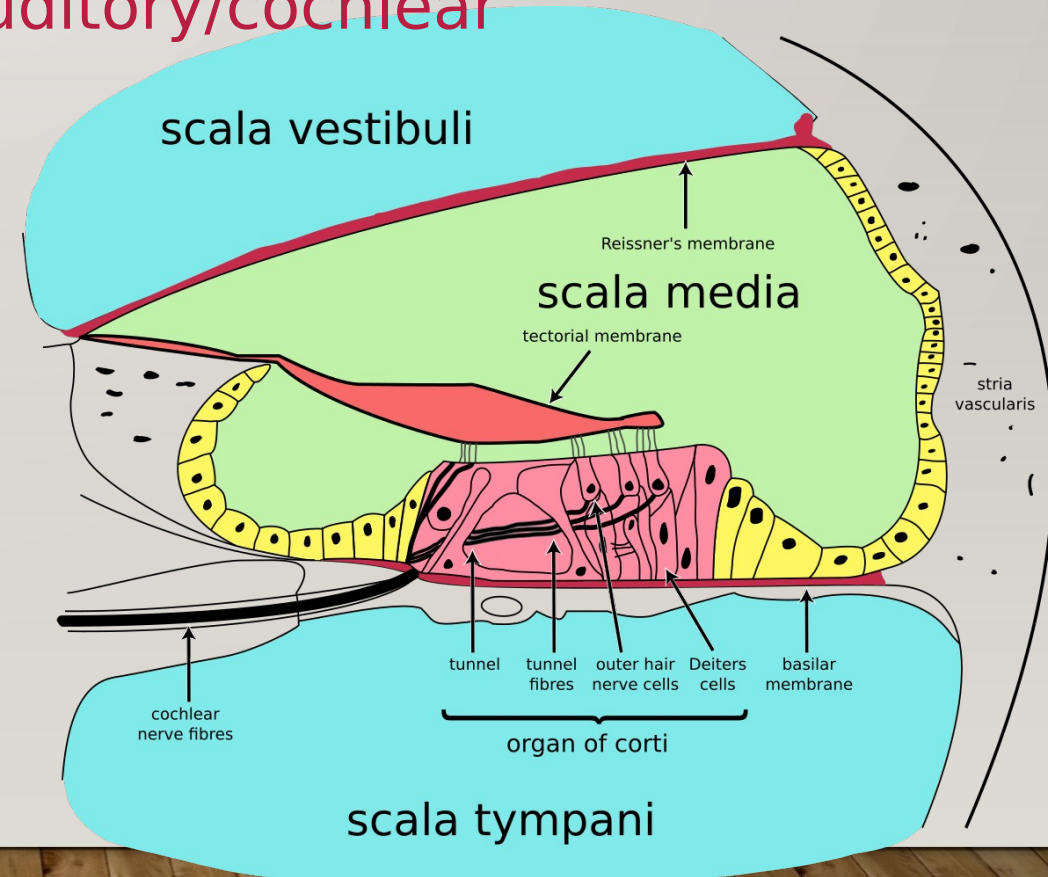
- **SOUND CONDUCTION**

- Pinna captures sound waves
- **tympanic membrane vibrates ossicles** (malleus, incus, stapes) – **these amplify sound** - vibrates oval window- moves fluid in inner ear which stimulates nerve receptors.





Cochlea hairs depolarize which converts the sound waves into electrical signals sent to the auditory cortex via the auditory/cochlear nerve.



COCHLEA AND HEARING

Several good videos to help you understand hearing!

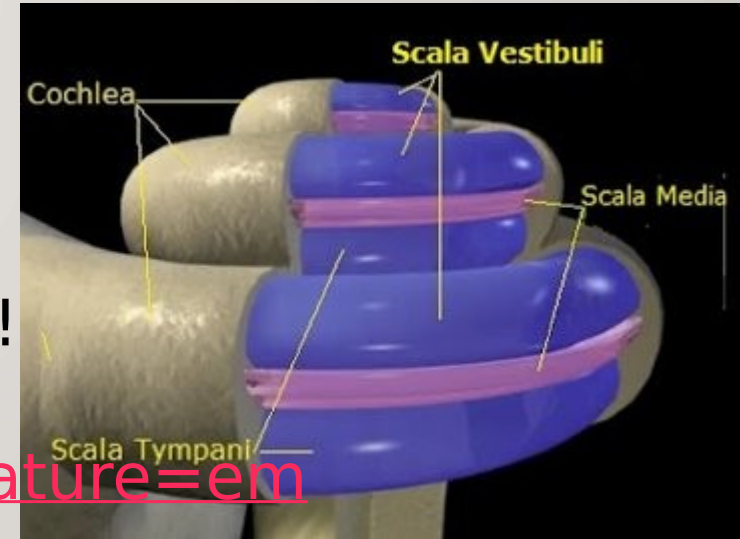
https://www.youtube.com/watch?v=PeTriGTENoc&feature=emb_logo

<https://www.youtube.com/watch?v=o2ItcKBMqGc>

https://www.youtube.com/watch?v=P5gyQf2gVvA&feature=emb_title

<https://www.youtube.com/watch?v=fIIAxGsV1q0>

<https://www.youtube.com/watch?v=le2j7GpC4JU> both
hearing and balance!



PRACTICE QUESTION

- Timothy is on a rollercoaster. Timothy does not like rollercoasters because he does not like the feeling of falling. He recently learned that there are calcium carbonate crystals in his ears that contribute to this sensation he dislikes. Which is true?

A – the crystals are called the cupula and are found in the semicircular canals

B – the crystals are called otoliths and are found in the utricle/sacculle

C – the crystals don't exist; Timothy is making things up

D – the crystals are called cochlear stones and are found in the basilar membrane

REMINDERS

DUE WEDNESDAY: 1st discussion post

DUE SUNDAY:

- Pre-lesson checkpoint
- Module 7 quiz (Respondus)
- Peer response in discussion
- Lab quiz (Respondus)

You can avoid feeling overwhelmed on Sunday by completing something each day – this also leads to better retention of knowledge.

Struggling?

Let's meet - send me an email and we can arrange a meeting time!